Between Visits

No symptom waits for the next visit, in any language, at any pace.

A voice agent calls cancer-trial patients in the days after a dose, in English or Spanish and at their pace. Fixed CTCAE rules grade the answers, the on-call nurse is paged for red flags, and the study team gets drafts with the patient's own words. On the sponsor side, **Site Signal** points a monitor at sites whose AE log disagrees with what their own patients reported.

The AI listens, rules grade, humans decide.

The patients trials struggle to reach between visits are the ones who don't speak English and the ones who need more time.

Non-English speakers

- English-only staff, apps and forms miss them.
- Clinical questions must not be machine-translated live.
- Between Visits asks pre-written, approved questions in English or Spanish.

Patients who need more time

- Fast voice bots and small screens fail older patients and people with low vision or literacy.
- The extra-time variants speak slower, wait longer and read every answer back.
- A plain phone-style conversation also catches what forms miss.

Source: Regeneron mentor conversation, Oct 10 (README § Why)

Rosa is 74, speaks Spanish and had her second dose two days ago. Her next visit is in three weeks.

What happens in the gap

- Side effects start between visits and surface at the next one, from memory.
- With an anti-PD-1 antibody, diarrhea can be the first sign of immune-related colitis; the example protocol treats 7 or more stools a day above baseline as grade 3.
- Serious events must reach the sponsor within 24 hours of site awareness.

Clinician reports miss patient-reported symptoms

- Across three randomized trials, physicians did not report 40.7–74.4% of the symptomatic toxicities patients reported (Di Maio 2015; secondary summary, UNVERIFIED against the full text).
- NCI built PRO-CTCAE for this reason (NCI).
- The patient stream is a reference, not the truth: patients report earlier and more often, clinicians judge attribution (Basch 2009; UNVERIFIED).

Sources: docs/10-site-signal.md §2 [1], [3], [5]; README § Why

HOW IT WORKS

Plan → call → nurse → review → sponsor

Plan

The study team approves plain-language questions in English and Spanish, red flags from the drug class's patient guide, and calls on days 1, 3 and 7 after each dose.

→

Call

The agent says it is an AI, offers a person, and asks only the question the backend hands it, in the patient's language and pace.

→

Nurse

Fixed CTCAE rules grade; a red flag pages the on-call nurse before the agent's next word, and re-pages until acknowledged.

→

Review

Drafts carry the verbatim quote, language, second and source; AI translations are labeled. A person confirms each one.

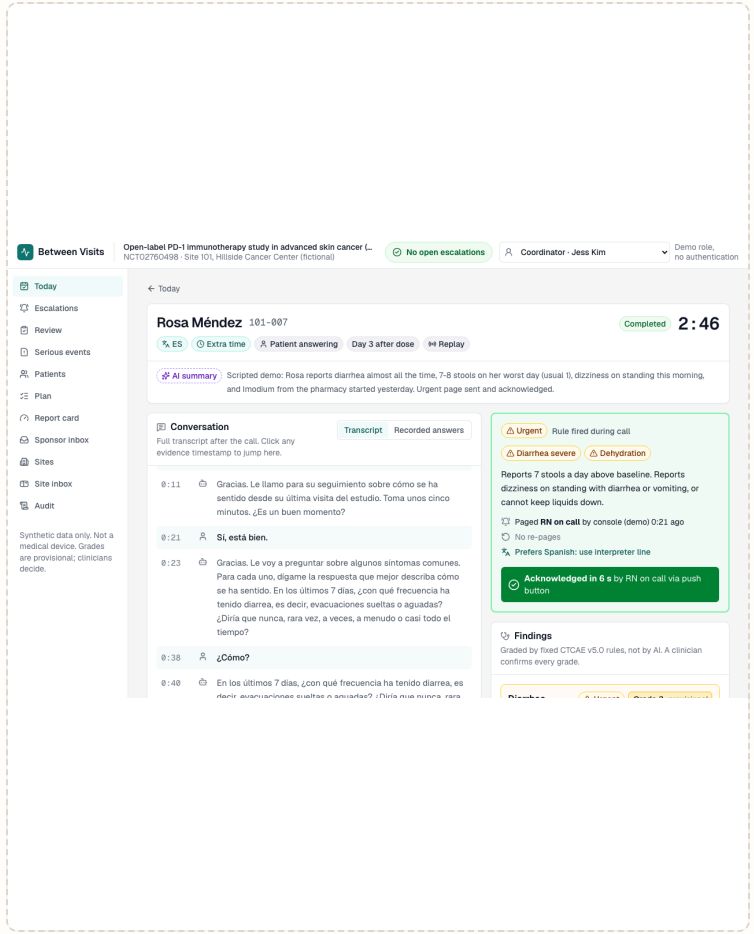
→

Sponsor

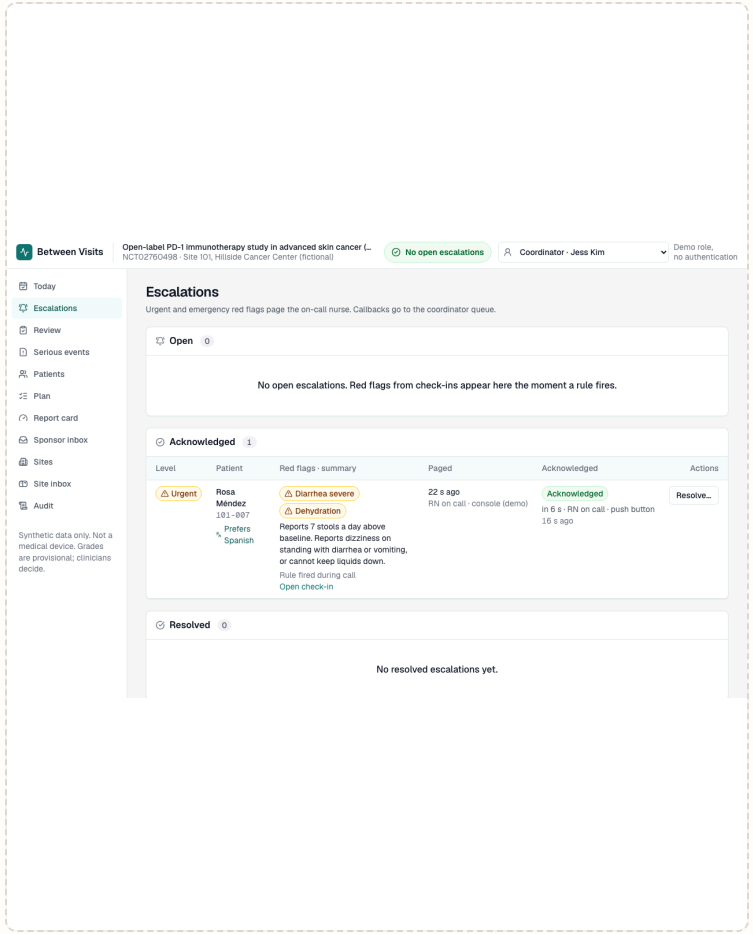
The investigator signs a serious-event report; the sponsor inbox shows the 24-hour clock, subject ID only.

Site Signal (sponsor side): each site's AE log is compared with what its own patients reported on these calls, relative to the other sites. A flag is a reason to look, not a finding.

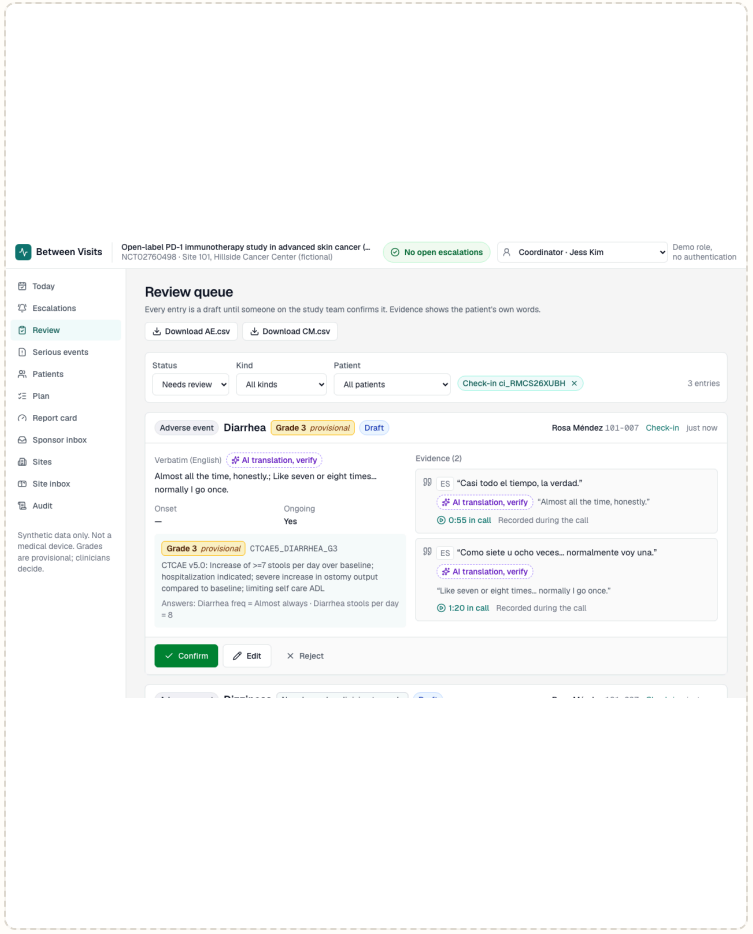
Rosa's call, the nurse page, and drafts with evidence



Live check-in: Diarrhea, **provisional** grade 3 by a fixed rule; urgent page sent.



The nurse acknowledges with one tap; the page carries the subject ID only.



Drafts with Rosa's Spanish words, the second, the source and **AI TRANSLATION**.

Synthetic patient. The call is labeled on screen: a real call played by a teammate, or "Scripted replay".

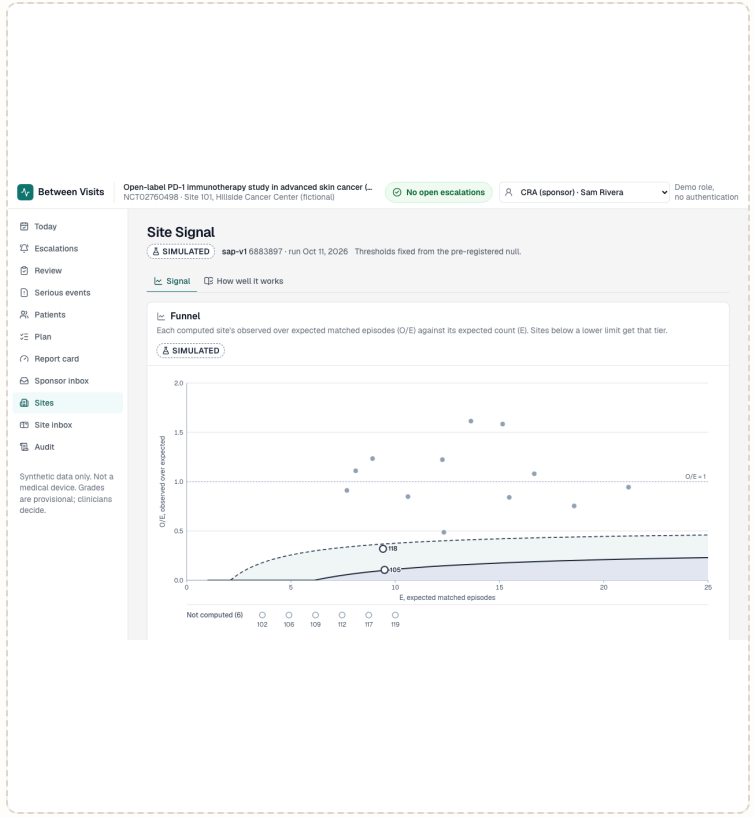
Each rule is a tested requirement

Rule	How it holds	Proof
Discloses AI, offers a human	First sentence of every call, every language	REQ-001, REQ-002
Approved words only	Questions and escalation lines come from data files in the call's language; no live translation of clinical text	REQ-003
No medical advice	Medical questions get the approved deferral line and become a nurse task	REQ-007, judge over every transcript
Rules grade, not the AI	Grades and escalation come only from fixed CTCAE v5 and red-flag rules	REQ-004, REQ-005
Fail safe	Errors on a red-flag path escalate to a person; paging never waits on statistics	backend tests
A human decides	Every confirmation, signature and site query is a person's action, audited in a hash chain	REQ-010, REQ-019

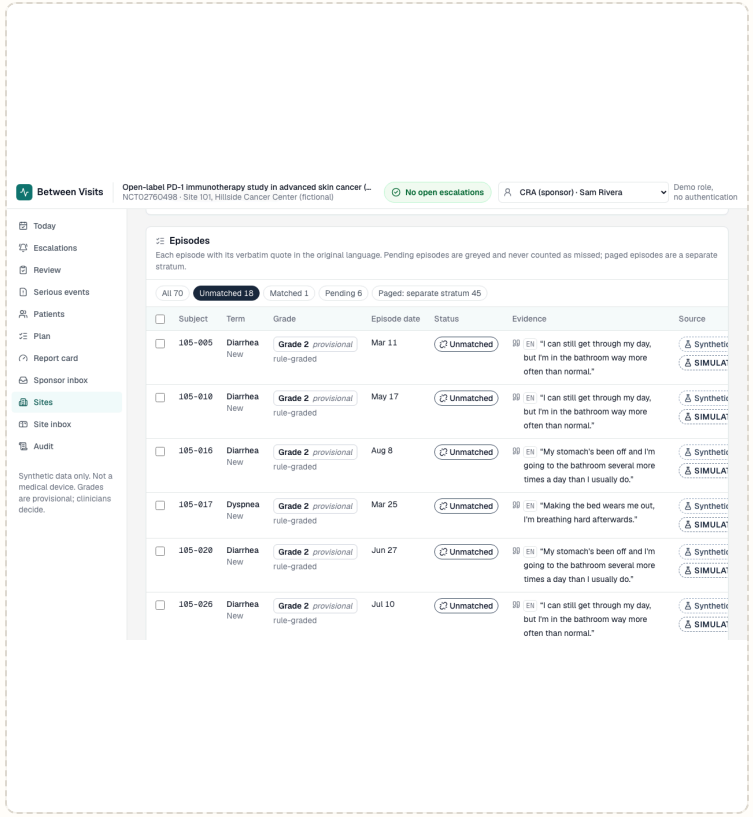
Real browser calls are reported as **MEASURED** in their own row (`eval/results/voice_checks.json`), never pooled with simulated results.

SITE SIGNAL **SIMULATED**

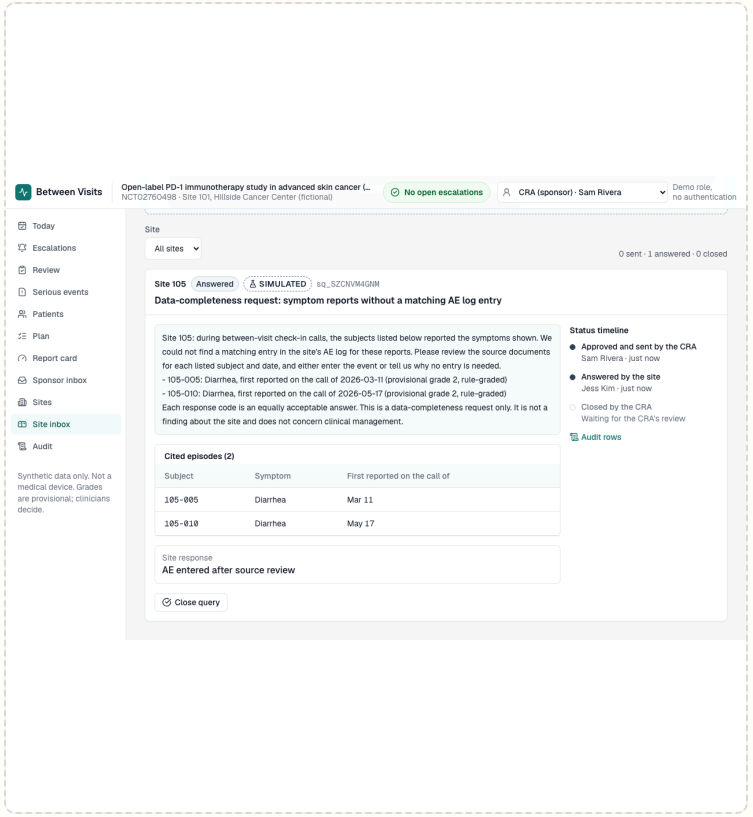
Which sites' AE logs disagree with what their own patients reported?



Funnel of observed vs expected capture for 14 computed sites. About 0.8 false Look flags are expected in a study with no problems.



A flag opens the unmatched episodes with each patient's verbatim words and provisional, rule-graded grade.



The CRA approves every templated query; "recorded as medical history" is a normal answer.

Capture discordance: episodes patients reported with no matching AE-log entry, compared with the other sites for the same symptom mix. A reason to look, not a finding. Thresholds were frozen in git (tag `sap-v1`) before any result existed.

Numbers: monitoring/out/signal_run.json, monitoring/out/thresholds.json (SIMULATED 20-site study)

Every number with its uncertainty, including what it misses

Between Visits report card SIMULATED PATIENTS, TEXT MODE

Urgent cases paged	**8 of 8** [0.631, 1]
Personas passing	19 of 20 [0.751, 0.999]
Expected side effects captured	1.0 [0.815, 1]
Medical advice given	0 flags

Known miss: P19 (Spanish, extra time), a false urgent page from the post-call safety net. 8 of 8 cannot rule out missing a third of red flags; 59 straight catches would show 95% recall.

Site Signal SIMULATED

Planted capture loss	Look or Watch	Watch
25%	0.15 ± 0.021	0.03 ± 0.009
50%	0.51 ± 0.029	0.22 ± 0.024
Gross (90%)	0.77 ± 0.024	0.70 ± 0.026
No planted site: false Look per site	0.05 ± 0.001 (nominal 0.05)	
No planted site: any false Watch per study	0.11 ± 0.007 (nominal 0.10)	

Not ahead everywhere: a plain AE-rate KRI flags the gross loss at Look or Watch in 0.99 ± 0.005; discordance needs enough episodes and computed the planted site in 0.77 of studies. Failure case (honest sites differ mainly in recording diligence), 50% loss, Look or Watch: discordance 0.30 ± 0.026 vs plain AE-rate KRI 0.82 ± 0.022. ± is the Monte Carlo SE.

eval/results/latest.json (overall.*, *_ci95: Clopper-Pearson 95%) · validation/site_signal/summary.json · WHAT_IT_GETS_WRONG.md

Runs from a fresh clone, and every claim has a check

One command

`make setup && make seed && make demo`: demo mode replays a call through the real backend with no API keys. MIT licensed.

CI

Backend tests, data validation with a product-name guard, frontend typecheck and lint, a quickstart job, and a full-history secret scan.

Traceability

Every requirement maps to tagged tests and their latest result (`validation/traceability.md`).

Pre-registered

The Site Signal analysis plan was tagged `sap-v1` before any output; the evaluation seed comes from a public beacon drawn after the tag.

Audit hash chain

Every human action is an append-only audit event; `GET /api/audit/verify` detects tampering inside the database.

Honest limits

`WHAT_IT_GETS_WRONG.md` and `NOT_BUILT.md`, one GitHub issue per item not built.

From simulation to a real trial

Validate Site Signal on real data

Retrospective analysis of completed trials that collected patient-reported and clinician-reported (CTCAE) symptoms for the same patients, with site IDs. Pre-registered as a new plan version before data access. This is the only test that does not depend on our simulator.

Pilot with one site

- Real voice calls with measured speech-recognition error by language and pace.
- More languages, professionally translated and clinically reviewed.
- EDC and FHIR integration; phone calls through Twilio.
- Real authentication, production hardening and regulatory review.

The AI listens, rules grade, humans decide.